Using MOI Transformation Equations to Compute I_u, I_v, I_{uv} at some angle θ from x.

Equations: Given or find for an Area (about its x and y axes): Ix, Iy, and Ixy

(Note: If I_{xy} = 0, then this Area's I_x and I_y are principal axes (I_{MAX} and I_{MIN})).

To find Iu, Iv and Iuv about an arbitrary rotated coordinate system (at angle θ):

1. Equations in terms of θ.
(Can use the Iu equation for both I_θ and $I_{\theta+90}$, just put the correct angle in the equation; so technically do not need the Iv eqn)

1(a) $$\boxed{I_u = I_x cos^2\theta + I_y sin^2\theta - 2I_{xy}\sin\theta\cos\theta}$$

1(b) $$\boxed{I_v = I_x sin^2\theta + I_y cos^2\theta + 2I_{xy}\sin\theta\cos\theta}$$

1(c) $$\boxed{I_{uv} = I_{xy}(cos^2\theta - sin^2\theta) + \left(I_x - I_y\right)sin\theta cos\theta}$$

For Iuv easier to use the 2θ form (2(c)) of the equation below.

2. Equations in terms of 2θ.
(Slightly easier to calculate, no need to square sin and cos terms)

2(a) $$\boxed{I_u = \frac{I_x + I_y}{2} + \frac{I_x - I_y}{2}cos2\theta - I_{xy}sin2\theta}$$

2(b) $$\boxed{I_v = \frac{I_x + I_y}{2} - \frac{I_x - I_y}{2}cos2\theta + I_{xy}sin2\theta}$$

2(c) $$\boxed{I_{uv} = \frac{I_x - I_y}{2}sin2\theta + I_{xy}cos2\theta}$$

Original Shape, I_x, I_y, I_{xy}, I_{AVG}, $(I_x - I_y)/2$,

Using the θ equations to find I_u, I_v, I_{uv} at θ = _______________.

For Iuv easier to use the 2θ form (2(c)) of the equation below.

2. Equations in terms of 2θ.
(Slightly easier to calculate, no need to square sin and cos terms)

2(a) $$\boxed{I_u = \frac{I_x + I_y}{2} + \frac{I_x - I_y}{2} cos2\theta - I_{xy} sin2\theta}$$

2(b) $$\boxed{I_v = \frac{I_x + I_y}{2} - \frac{I_x - I_y}{2} cos2\theta + I_{xy} sin2\theta}$$

2(c) $$\boxed{I_{uv} = \frac{I_x - I_y}{2} sin2\theta + I_{xy} cos2\theta}$$

Using the 2θ equations to find I_u, I_v, I_{uv} at θ = ____________, 2θ = ______________.